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**Java SE 27 & JDK 27**

# Class CoderResult

java.lang.Object
    java.nio.charset.CoderResult

---

```
public final class CoderResult
extends Object
```

A description of the result state of a coder.

A charset coder, that is, either a decoder or an encoder, consumes bytes (or characters) from an input buffer, translates them, and writes the resulting characters (or bytes) to an output buffer. A coding process terminates for one of four categories of reasons, which are described by instances of this class:

- *Underflow* is reported when there is no more input to be processed, or there is insufficient input and additional input is required. This condition is represented by the unique result object `UNDERFLOW`, whose `isUnderflow` method returns `true`.
- *Overflow* is reported when there is insufficient room remaining in the output buffer. This condition is represented by the unique result object `OVERFLOW`, whose `isOverflow` method returns `true`.
- A *malformed-input error* is reported when a sequence of input units is not well-formed. Such errors are described by instances of this class whose `isMalformed` method returns `true` and whose `length` method returns the length of the malformed sequence. There is one unique instance of this class for all malformed-input errors of a given length.
- An *unmappable-character error* is reported when a sequence of input units denotes a character that cannot be represented in the output charset. Such errors are described by instances of this class whose `isUnmappable` method returns `true` and whose `length` method returns the length of the input sequence denoting the unmappable character. There is one unique instance of this class for all unmappable-character errors of a given length.

For convenience, the `isError` method returns `true` for result objects that describe malformed-input and unmappable-character errors but `false` for those that describe underflow or overflow conditions.

**Since:**

1.4

## *Field Summary*

**Fields**

| Modifier and Type | Field | Description |
| --- | --- | --- |
| static final CoderResult | OVERFLOW | Result object indicating overflow, meaning that there is insufficient room in the output buffer. |
| static final CoderResult | UNDERFLOW | Result object indicating underflow, meaning that either the input buffer has been completely consumed or, if the input buffer is not yet empty, that additional input is required. |

## *Method Summary*

All Methods | Static Methods

| Modifier and Type | Method | Description |
| --- | --- | --- |
| boolean | isError() | Tells whether or not this object describes an error condition. |
| boolean | isMalformed() | Tells whether or not this object describes a malformed-input error. |
| boolean | isOverflow() | Tells whether or not this object describes an overflow condition. |
| boolean | isUnderflow() | Tells whether or not this object describes an underflow condition. |
| boolean | isUnmappable() | Tells whether or not this object describes an unmappable-character error. |
| int | length() | Returns the length of the erroneous input described |

| | | |
|---|---|---|
| | | by this object *(optional operation)*. |
| static CoderResult | **malformedForLength**(int length) | Static factory method that returns the unique object describing a malformed-input error of the given length. |
| void | **throwException**() | Throws an exception appropriate to the result described by this object. |
| String | **toString**() | Returns a string describing this coder result. |
| static CoderResult | **unmappableForLength**(int length) | Static factory method that returns the unique result object describing an unmappable-character error of the given length. |

**Methods declared in class Object**

clone, equals, finalize, getClass, hashCode, notify, notifyAll, wait, wait, wait

## *Field Details*

### UNDERFLOW

```
public static final CoderResult UNDERFLOW
```

Result object indicating underflow, meaning that either the input buffer has been completely consumed or, if the input buffer is not yet empty, that additional input is required.

### OVERFLOW

```
public static final CoderResult OVERFLOW
```

Result object indicating overflow, meaning that there is insufficient room in the output buffer.

## *Method Details*

### toString

`public String toString()`

Returns a string describing this coder result.

**Overrides:**

`toString` in class `Object`

**Returns:**

A descriptive string

### isUnderflow

`public boolean isUnderflow()`

Tells whether or not this object describes an underflow condition.

**Returns:**

`true` if, and only if, this object denotes underflow

### isOverflow

`public boolean isOverflow()`

Tells whether or not this object describes an overflow condition.

**Returns:**

`true` if, and only if, this object denotes overflow

### isError

`public boolean isError()`

Tells whether or not this object describes an error condition.

**Returns:**

`true` if, and only if, this object denotes either a malformed-input error or an unmappable-character error

### isMalformed

`public boolean isMalformed()`

Tells whether or not this object describes a malformed-input error.

**Returns:**

`true` if, and only if, this object denotes a malformed-input error

## isUnmappable

```
public boolean isUnmappable()
```

Tells whether or not this object describes an unmappable-character error.

**Returns:**

`true` if, and only if, this object denotes an unmappable-character error

## length

```
public int length()
```

Returns the length of the erroneous input described by this object *(optional operation)*.

**Returns:**

The length of the erroneous input, a positive integer

**Throws:**

`UnsupportedOperationException` - If this object does not describe an error condition, that is, if the `isError` does not return `true`

## malformedForLength

```
public static CoderResult malformedForLength(int length)
```

Static factory method that returns the unique object describing a malformed-input error of the given length.

**Parameters:**

`length` - The given length

**Returns:**

The requested coder-result object

## unmappableForLength

```
public static CoderResult unmappableForLength(int length)
```

Static factory method that returns the unique result object describing an unmappable-character error of the given length.

**Parameters:**

`length` - The given length

**Returns:**

The requested coder-result object

### throwException

```
public void throwException()
                    throws CharacterCodingException
```

Throws an exception appropriate to the result described by this object.

**Throws:**

`BufferUnderflowException` - If this object is `UNDERFLOW`

`BufferOverflowException` - If this object is `OVERFLOW`

`MalformedInputException` - If this object represents a malformed-input error; the exception's length value will be that of this object

`UnmappableCharacterException` - If this object represents an unmappable-character error; the exception's length value will be that of this object

`CharacterCodingException` - `MalformedInputException` if this object represents a malformed-input error; `UnmappableCharacterException` if this object represents an unmappable-character error

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Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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